30-Minute Classroom Task (Non-Coding)

Topic 1: Strassen’s Matrix Multiplication

Task 1 (5 Minutes)

Answer the following:

1. What is a Matrix?
2. What is the full name of Strassen’s Algorithm?
3. Why was Strassen’s Algorithm developed?
4. How many multiplications are required in Normal Matrix Multiplication for a 2×2 matrix?
5. How many multiplications are required in Strassen’s Algorithm?

Ans: 1. arrangement of rows and columns

2. Strassen’s Matrix Multiplication Algorithm

3. For faster multiplication of larger matrix

4. 8

5. 7

Task 2 (5 Minutes)

Fill in the blanks:

1. Strassen’s Algorithm uses merge and conquer.
2. A matrix is arranged in rows and columns.
3. Strassen’s Algorithm reduces multiplications from 8 to 7.
4. The final matrix contains values C11, C12, C21 and C22.
5. Strassen’s Algorithm is faster for large matrices.

Topic 2: Recurrence Relations

Task 3 (5 Minutes)

Write True or False:

1. Recurrence Relations are used to analyze recursive algorithms. True

2. Binary Search recurrence is $T(n)=T(n-1)+1$. True
3. Fibonacci depends on previous values. True
4. Merge Sort uses Divide and Conquer. True
5. Recurrence Relations describe a problem using smaller versions of itself. True

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Task 4 (5 Minutes)

Match the Following:

Ans: a- 1 b-2,c-3

Algorithm	Recurrence Relation
Factorial	$T(n)=2T(n/2)+n$
Binary Search	$T(n)=T(n-1)+1$
Merge Sort	$T(n)=T(n/2)+1$

Topic 3: Time Complexity Analysis

Task 5 (5 Minutes)

Write the Time Complexity:

1. Linear Search → 0(n)
2. Binary Search → O(log n)
3. Bubble Sort → 0(n^2)
4. Merge Sort → O(nlog n)
5. Strassen’s Algorithm → **O(n^(log 7))**.

Task 6 (5 Minutes)

Short Answers:

1. What is Time Complexity? It describes how running time of an algorithm increases with increase in the input.
2. Why is Binary Search faster than Linear Search? Because it eliminates half of the data in each iteration,
3. Which is better: O(n) or $O(n^2)$? O(n) because it grows proportionally with the input
4. What does O(log n) mean? As n increases time complexity increases at a slow rate
5. Why do we analyze algorithms? On the basics of time and space complexity

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Bonus Question

Which algorithm is faster for large matrices?

A) Normal Matrix Multiplication
B) Strassen's Matrix Multiplication

Explain your answer in 2 lines.

B because it reduces multiplications from 7 to 8 and its good for large matrices.